The Impact of Post-Training Techniques on Feature Representations

Alberto Puliga
BCSAI - IE University

Deployment outpaces understanding

High-stakes

LLMs are now deployed in medicine, law, financial institutions, military, etc.

Black Boxes

We can evaluate outputs, but not the internal computations behind them.

Deeper problem

Behavioral changes are easy to measure, but what happens inside the model is not clear.

Training Pipeline

Pre-training
(knowledge)

Learn broad language patterns from internet-scale text

Post-training
(behaviour)

Align behavior with supervision and preference optimization

Final model

Final assistant optimized to be helpful, honest, and harmless

Mechanistic Interpretability

Goal

Reverse-engineer neural networks into human-understandable algorithms.

Polysemanticity

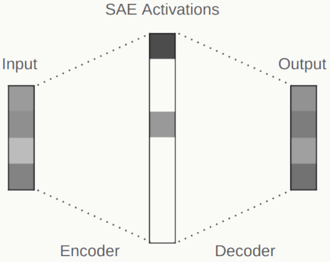
Individual neurons activate for multiple and unrelated concepts.

$\#$ of features $>$ model dimensions

Unit of analysis

Sparse Autoencoder (SAEs)

An SAE learns a wide dictionary of directions over a model's activations. A sparse encoder maps each activation to only a handful of active entries; the decoder reconstructs the original. Sparsity forces each dictionary entry to be monosemantic.



This gives us a feature-level vocabulary for the residual stream

Building on Bricken et al. (2023) and Templeton et al. (2024)

How do the features learned during pre-training change after post-training,

and how do they relate to behavioural changes?

Hypothesis

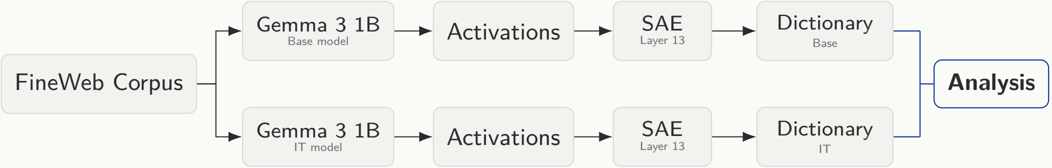
Empirical claim

Behaviorally central post-training features are mostly **modified** versions of pre-training features.

"Modified" = matched either by decoder cosine similarity or activation correlation across checkpoints, but with shifted direction or function.

Methodology

4 Stage Pipeline



Methodology

Analysis

1. Classify

4 Classes

- Preserved
- Modified
- PT-exclusive
- IT-exclusive

τ_d = decoder cosine sim

τ_a = activation correlation

2. Label

200 features sampled
by activation frequency

Dual-LLM consensus

Claude Haiku 4.5
GPT-5.4-nano

6 semantic categories

e.g. Syntax, Instruction, Safety...

3. Probe

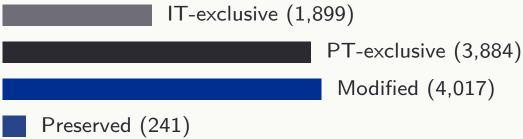
Welch t-tests per feature

Safety contrast
JBB harmful vs benign

Instruction contrast
Alpaca vs TruthfulQA

Which class do the behaviourally relevant features belong to?

Result 1: Modified Features Lead



25 threshold pairs tested

Interpretation: post-training mostly reshapes pre-training features, rather than creating entirely new ones.

Result 2: Instruction-Following Signal Is Shared

19/20

top IT features labeled

instruction_style

Shared top indices in PT and IT

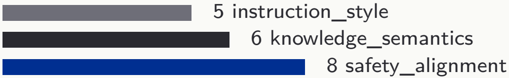
2006 446

In PT these indices still show large effects (+114.3, +37.5).
They belong to the **modified** class, not IT-exclusive.

Interpretation: amplification and sharpening,
not de novo creation.

Result 3: Safety Features Also Overlap

IT Top-20 safety labels



Shared high-rank indices

262 2184

No IT top-20 safety feature is IT-exclusive. Safety shifts are real, but mostly on shared PT substrate.

Limitations

Correlational evidence

The analysis links feature shifts to behavior, but it does not yet prove that those features causally drive the behavior.

Narrow experimental scope

Results come from one matched PT/IT model pair, one architecture scale, and one layer rather than a broad cross-model study.

Imperfect semantic labeling

Feature taxonomy is useful but noisy: labels are partly interpretive, and inter-annotator agreement reached about 60%.

Conclusion

Main takeaway

Post-training appears to act less by creating entirely new internal structure and more by **reorganizing, amplifying, and sharpening existing pre-training features**.

- Modified features dominate the matched dictionaries, while preserved features remain a small minority.
- The strongest instruction-following and safety-related IT features mostly trace back to shared PT substrate rather than to purely IT-exclusive features.
- The evidence therefore supports a **weak but consistent claim**: behaviorally central post-training features are usually transformed versions of pre-training features.

Next Steps

1. Test causality

Intervene on high-impact modified features and check whether steering them changes refusal or instruction-following behavior.

2. Expand the scope

Replicate the pipeline across more layers, larger model sizes, and different model families to test generality.

3. Improve measurement

Refine the labeling scheme and add richer behavioral probes so the feature-behavior link is measured more precisely.